Lab 7 – Prompts 03 & 04: Literature-to-Experiment Bridge and Novelty/Feasibility Decision

Oct 23, 2026 · @Vibhum Sharma

Outcome: conditional GO for a controlled comparison (Pathway B) with a narrowed claim. The search found close prior work: Kim et al. (MLHC 2025) already tested specialist roles and discussion vs voting on 31 private oncology cases. What remains untested is a **call-matched** comparison on **public** oncology questions with an **open 7B model**, plus how often discussion turns correct answers wrong.

Item	Detail
Course	AI Tools for Research, Lab 7 (post-mid-sem), Phase I Prompts 03–04
Student	Vibhum Sharma (Topic ID T32)
Inputs	Lab 4 review; Prompt 02 gap matrix (G4 ranked first)
AI tool	Claude, with web search and a Linux workspace (no GPU)

Labels: **VERIFIED** = read in the original source or confirmed by running something. **INFERRED** = plausible, or taken from an AI summary of a source I have not read myself. **UNKNOWN** = no evidence yet.

Prompt 03 — Map the literature to an executable experiment

Prompt as used (brackets filled from the Prompt 02 output and my own constraints):

> My literature review **[LiteratureReview_T32_VibhumSharma.pdf]** identified these candidate gaps: **[G4 multi-agent claims lack independent, fair comparison; G1 complexity gap; G2 answer instability]**. Identify three published methods from my review that could realistically be implemented or reproduced. For each method, map its original paper to the algorithm, dataset, code or model access, evaluation metric, known limitations and computational requirement. Distinguish verified access to code/weights from unverified suggestions. Compare three possible experimental studies: (A) reproduction plus extension, (B) controlled comparison, and (C) targeted modification. For each study state the smallest working baseline, the testable question, expected resource burden and evidence required. Recommend one minimum viable research experiment for **[free Kaggle 2×T4 / Colab T4, local GTX 1050 Ti 4 GB, 16 GB RAM]** and **[about 4 weeks]**. Provide a literature-to-experiment mapping table and list what I must personally verify before coding. Do not propose an elaborate new architecture without a scientific rationale.

Output A — Three methods from the review, mapped to what is actually reproducible

None of the three multi-agent systems in my review can be reproduced as published on free GPUs. EvoMDT's code link did not resolve, Han et al. name no code or models, and the single-LLM studies use private cases. So the experiment must re-implement the shared core idea (specialist agents + discussion + aggregation) rather than reproduce one system.

	M1. EvoMDT (Liu et al. 2026)	M2. Consensus matrix (Han et al. 2025)	M3. Guideline-primed single LLM (Ismayilov et al. 2025)
Algorithm	Specialist agents, guideline RAG, safety agent, coordinator consensus, self-evolving prompts (VERIFIED, PDF)	7 role agents; agreement measured by Kendall's W; RL (Q-learning, PPO, DQN) to improve consensus (VERIFIED, arXiv abstract)	One LLM primed with NCI guidelines simulates the board (review text only; original not read — UNKNOWN)

	M1. EvoMDT (Liu et al. 2026)	M2. Consensus matrix (Han et al. 2025)	M3. Guideline-primed single LLM (Ismayilov et al. 2025)
Data	MedQA, MedQA-Cancer, MedMCQA-Cancer, MedQuAD, MedQARo, ROND + 4 hospital sets "on request" (VERIFIED)	MedQA, PubMedQA, DDXPlus, MedBullets, SymCat (VERIFIED, abstract)	10 NSCLC immunotherapy cases, private (review)
Code / weights	Paper links github.com/KesselZ/EvoMDT; **could not be opened** from here (UNKNOWN). Backbone DeepSeek V3/R1 via API (VERIFIED)	No code or model named (UNKNOWN)	Prompt only; no code (review)
Metric	Accuracy (MCQ), BERTScore, physician Likert ratings (VERIFIED)	Accuracy, consensus rate, Kendall's W, expert rating (VERIFIED)	Likert 1–5 for accuracy and consistency (review)
Known limits	Retrospective; limited scale and geography; no prospective or outcome study (VERIFIED)	Strongest baseline not named (VERIFIED)	No insight beyond the human MDT (review)
Compute	API calls to DeepSeek V3/R1, far too large to run on a T4	Many agents + RL training: high	Trivial

Open implementations outside my review, usable as references: MedAgents (Tang et al. 2024), MDAgents (Kim et al. 2024) and Kim et al. (2025) all have public GitHub code (VERIFIED). All three call the OpenAI or Gemini APIs only, and none has a licence file (VERIFIED). No licence means I can read and cite them but should not copy their code, so the baseline is my own implementation.

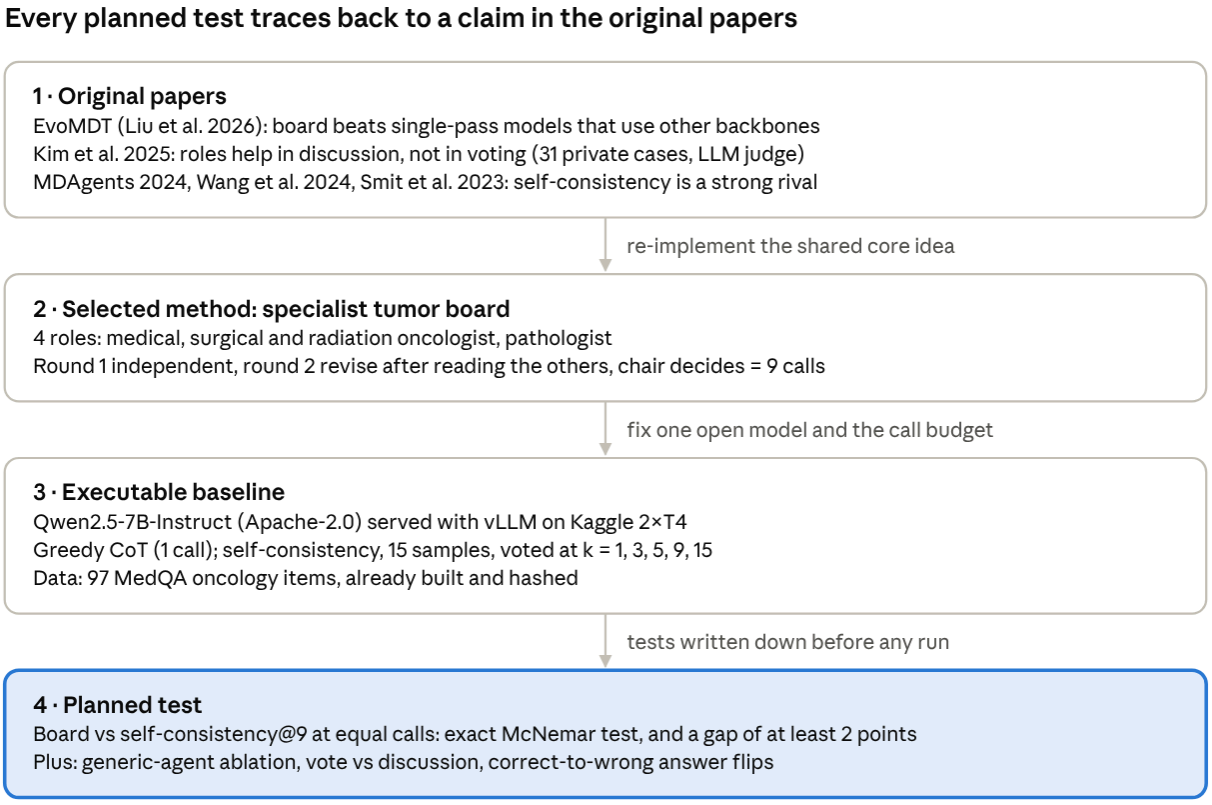
Output B — Three possible studies compared

Study B is the only one with a working baseline I already control, a clean test, and a compute bill that fits free GPUs.

	A. Reproduction + extension	B. Controlled comparison	C. Targeted modification
What	Reproduce EvoMDT (or Kim et al.'s Group Chat), then add a new condition	Fix one open model; compare single-model and board methods under equal calls	Change one part of a board, e.g. discuss only when round-1 members disagree
Smallest working baseline	EvoMDT pipeline + its MCQ sets	Greedy CoT (1 call) and self-consistency@k on MedQA oncology items with Qwen2.5-7B	The full board from study B
Testable question	Does the published gain hold on independent data?	Does a specialist board beat self-consistency at the same call count? Do roles matter?	Does the change keep accuracy with fewer calls?
Resource burden	High: code unreachable, API model, private data	Medium: about 34 calls per item, all inference (INFERRED estimate)	Medium, but only after B exists
Evidence required	Matching the published numbers first	Paired accuracy, McNemar, CIs, tokens per item, flip counts	Accuracy vs calls curve against the unmodified board
Verdict	**Not feasible**	**Recommended**	Possible follow-up, not now

Recommended minimum viable experiment: study B on the 97 MedQA oncology items first (the subset is already built and hashed), with Qwen2.5-7B-Instruct on Kaggle 2×T4. It frames Kim et al. 2025's hypotheses as a conceptual replication on public data, with call matching as the one new control. No new architecture is proposed.

Output C — Mandatory bridge: original papers → selected method → executable baseline → planned test



mandatory bridge · 4 stages

Each planned comparison answers a specific claim above it. The board-vs-self-consistency test answers EvoMDT's untested confound and the MDAgents/Wang/Smit evidence. The ablations repeat Kim et al.'s role and discussion findings on public data.

Output D — What I must verify myself before coding

Seven checks remain. Two involve numbers the AI got from a summary, not from reading the paper itself.

- [ ] Open github.com/KesselZ/EvoMDT in my own browser; the AI's fetch and clone both failed
- [ ] Read EvoMDT Figure 2 for any DeepSeek-only (same-backbone) baseline
- [ ] Read MDAgents Table 2 and Section 4.4 myself (CoT-SC 83.9 vs MDAgents 88.7 came from an AI summary)
- [ ] Read MedAgentsBench's results table myself (numbers came from an AI summary)
- [ ] Read Kim et al. 2025 Appendix B tables and confirm their call counts are not matched
- [ ] Check my Kaggle account's weekly GPU quota
- [ ] Confirm the MedMCQA dataset card on Hugging Face (licence, fields, dev split size)

Prompt 04 — Check feasibility, novelty and resource access

Prompt as used:

> My candidate study is **[a controlled comparison of a 4-specialist LLM tumor board against call-matched self-consistency, same open backbone (Qwen2.5-7B-Instruct), on oncology items from MedQA, with a generic-agent ablation]**. Develop a feasibility and novelty-verification plan. Provide scholarly search strings and specific categories of neighboring work to inspect, without inventing titles, DOIs or URLs. Create a matrix with columns: published method, VERIFIED contribution (with source), limitation, proposed difference and evidence still missing. Check whether the dataset, code, model weights, license and benchmark protocol are accessible; flag all unresolved availability claims. Estimate computational and time constraints for **[Kaggle 2×T4 free tier; Colab T4; GTX 1050 Ti 4 GB]** and **[4 weeks]**. Narrow the study to a viable experimental question and recommend a fallback if the preferred repository, weights or dataset are unavailable.

Output A — Search strings and neighbouring work

The searches below were actually run on 20–23 Oct 2026 (Google-style web search and arXiv/ACL/NeurIPS pages). The last two columns record what each one turned up.

#	Search string	Category of neighbouring work	Relevant hit
S1	("multi-agent" OR "agent teams") AND ("tumor board" OR "multidisciplinary team") AND (LLM OR "large language model")	Oncology MDT simulations	Kim et al. 2025 (MLHC); Kuerbanjiang et al. 2025; Zhang et al. 2025 (GI oncology); CoMMa 2026
S2	"multi-agent debate" AND ("self-consistency" OR "majority voting") AND ("compute" OR "budget" OR "cost")	Compute-matched test-time scaling	Wunderlich et al. 2026 (ACL SRW); Smit et al. 2023
S3	"multi-agent" AND medical AND (MedQA OR benchmark) AND (CoT-SC OR baseline)	Medical multi-agent frameworks	MDAgents 2024; MedAgents 2024; MedAgentsBench 2025

#	Search string	Category of neighbouring work	Relevant hit
S4	(persona OR "role prompting" OR "role specialization") AND LLM AND (ablation OR "does not improve")	Role/persona effects	Zheng et al. 2024 (EMNLP Findings)
S5	"Are multi-agent discussions the key" / "single agent" strong prompt	Single agent vs discussion	Wang et al. 2024 (ACL)
S6	"consensus matrix" oncology MDT	Consensus aggregation	Han et al. 2025 (arXiv)

Not yet searched (to do before the paper): conformity and error propagation in LLM discussions; oncology subsets of MedQA used by other studies; Google Scholar "cited by" for Kim et al. 2025.

Output B — Novelty matrix

The broad question "is multi-agent better than self-consistency" is already studied, with mixed answers. Kim et al. 2025 already ran the oncology role ablation. The defensible difference is narrower: strict call matching, public data with an objective metric, an open 7B model, and spontaneous error propagation.

Published method	Contribution (source, label)	Limitation	My proposed difference	Evidence still missing
EvoMDT, Liu et al. 2026	MedQA 0.86 vs Llama3-70B 0.76, Claude-3 0.79; backbone DeepSeek V3/R1 (PDF, VERIFIED)	Baselines use other models, single pass; code link unreachable	Same backbone for every arm; equal call budget	Fig. 2 not checked for a DeepSeek-only baseline

Published method	Contribution (source, label)	Limitation	My proposed difference	Evidence still missing
Kim et al. 2025, MLHC (head and neck MDT)	Generic roles hurt Group Chat (win 25.8% vs 53.5%, p<0.001) but not Majority Vote (ns); adding discussion to voting helps; GPT-4o plus Gemma-27B, MedGemma-27B, o4-mini (PDF text, VERIFIED)	31 private single-centre cases; LLM-as-judge pairwise wins; call counts differ between methods; code has no licence	Public oncology MCQs with exact accuracy; call-matched; 7B open model; correct→wrong flips without injected errors	Whether their discussion gain survives call matching
MDAgents, Kim et al. 2024, NeurIPS	Has CoT-SC baseline on the same GPT-4 backbone: MedQA CoT-SC 83.9, MedAgents 79.1, MDAgents 88.7; API calls reported for 3 configs only (AI summary of PDF, INFERRED)	Closed model; calls not matched per method; no oncology subset	Oncology subset; matched calls; open model	My own reading of Table 2
MedAgentsBench, Tang et al. 2025	MedAgents and MDAgents average below CoT-SC on the HARD subset with GPT-4o; no matched-cost comparison (AI summary, INFERRED)	No oncology focus; GPT-4o only for agent methods	Matched calls; oncology	My own reading

Published method	Contribution (source, label)	Limitation	My proposed difference	Evidence still missing
Wang et al. 2024, ACL	Strong single-agent prompt ≈ best discussion; discussion wins only without demonstrations (abstract, VERIFIED)	General reasoning tasks, no medical roles	Medical oncology roles; zero-shot (no demonstrations)	—
Smit et al. 2023	Debate does not reliably beat self-consistency (abstract v3, VERIFIED)	Closed models	Open 7B; oncology	—
Zheng et al. 2024, EMNLP Findings	Personas in system prompts do not improve performance (title, VERIFIED)	Not multi-agent	Tests roles inside a board	Full-text check
Han et al. 2025, arXiv	87.5% vs 83.8% for an unnamed strongest baseline (abstract, VERIFIED)	Baseline and LLMs unnamed; no code	— (not reproducible)	Full text

Claim I can defend: an independent, call-matched, public-benchmark replication of the specialist-board idea with a small open model. That is a course-level contribution, not a new method.

Output C — Access and licence matrix

Everything the primary experiment needs is accessible and licensed for research. Three things are untested until the Kaggle run: the MedMCQA download, the model download, and vLLM on the Kaggle image.

Resource	Source	Licence	Access status
MedQA test, 5 options (1,273 items)	Jin et al. 2021, github.com/jind11/MedQA	MIT (LICENSE file, VERIFIED)	Downloaded via the MedAgents GitHub copy; sha256 recorded; 97 oncology items built (VERIFIED by execution)
MedMCQA dev	Pal et al. 2022, github.com/medmcqa/medmcqa; HF copy openlifescienceai/medmcqa	MIT (LICENSE file, VERIFIED)	Hugging Face is blocked from the AI workspace; download not tested (UNKNOWN)
Qwen2.5-7B-Instruct weights	huggingface.co/Qwen/Qwen2.5-7B-Instruct	Apache-2.0 (model card, VERIFIED)	Download on Kaggle not tested (UNKNOWN)
Inference engine	vLLM	Open source	Docs list T4 as supported, compute capability ⩾ 7.0 (VERIFIED, v0.9.0 docs); install on Kaggle untested (UNKNOWN)
Baseline code	My own implementation	MIT	Written; unit tests pass with a mock model (VERIFIED, 20 Oct); no real-model run yet
Reference code	MedAgents, MDAgents, Kim et al. 2025 on GitHub	**No licence file** (VERIFIED)	Read and cite only; do not copy
EvoMDT code	Link stated in paper	—	Could not be opened (UNKNOWN)
Benchmark protocol	Exact-match accuracy on multiple choice, as in MedAgents and MDAgents	—	Standard; no private test server

Output D — Compute, time, narrowed question and fallbacks

The minimum experiment should take roughly 1–3 GPU-hours and the full grid roughly 5–20. These are INFERRED estimates; the Kaggle smoke test will replace them with measured numbers.

Run	Items × seeds	Calls per item	Generated tokens (at ~250 per call)	GPU-hours on 2×T4
Minimum: MedQA oncology	97 × 3 = 291	34 (CoT 1 + SC 15 + board 9 + generic board 9)	about 2.5 M	about 1–3
+ MedMCQA oncology	n unknown × 3	34	unknown	scales with n
+ full MedQA (power set)	1,273 × 1	34	about 10.8 M	about 4–12

Assumptions (all INFERRED): 250 output tokens per call on average (cap 512); vLLM throughput of 500–1,500 generated tokens/s for a 7B fp16 model batched over two T4s; prompt processing adds up to 2×. Kaggle allows 12-hour sessions; its weekly GPU quota is not published and is commonly reported as about 30 hours (INFERRED). The local GTX 1050 Ti (4 GB) cannot hold a 7B model in fp16, so it is only for CPU tests and debugging.

Narrowed research question: with Qwen2.5-7B-Instruct fixed, on oncology items from MedQA, does a 4-specialist tumor board (9 calls) answer more accurately than self-consistency with 9 samples? Secondary: does replacing specialists with generic agents change board accuracy, and how often does round-2 discussion turn a correct member wrong?

If this fails	Fallback
vLLM will not install or run on Kaggle T4	Hugging Face transformers on one T4 (slower); cut to MedQA oncology, 1 seed
7B too slow or out of memory	Qwen2.5-3B-Instruct, logged as a protocol change
MedMCQA cannot be downloaded	MedQA oncology only, with full MedQA as the power set
GPU quota runs out	Drop full MedQA first, then seeds 3 → 1
Baseline accuracy near chance (5 options = 20%)	Stop and debug the prompt and parser on MedQA **train** items, never test items

Go/no-go decision

GO, conditionally, for study B with the narrowed claim. The question is answerable with public, licensed data and an open model, and it fits free GPUs. The novelty is modest and has to be stated honestly: it extends Kim et al. 2025 and MDAgents rather than opening a new area.

Condition	Kills or changes the plan if…
Kaggle smoke test (5 pilot items) runs end to end	vLLM and the HF fallback both fail → switch to the 3B model
Parse failures ⩽ 3% per method on pilot items	Higher → fix the prompt on train items before any test run
CoT accuracy on MedQA oncology clearly above 20%	Near chance → no-go for 7B; reassess
I read Kim et al. 2025 and MDAgents Table 2 myself and confirm no call-matched oncology comparison exists	One exists → reframe as pure replication (pathway A)
Measured GPU-hours fit my quota	Over budget → apply fallbacks in the order above

Decision log and AI-use record

research_decisions.md — entries added during Prompts 03–04

ID	Date	Decision	Reason	Evidence (label)	Next action
L-06	23 Oct 2026	Do not reproduce EvoMDT or Han et al.	Code unreachable or absent; API-only backbone; private data	Access matrix (VERIFIED / UNKNOWN as marked)	—
L-07	23 Oct 2026	Choose study B (controlled comparison) on MedQA oncology items	Only option with a baseline I control that fits free GPUs	Output B of Prompt 03	Kaggle smoke test
L-08	23 Oct 2026	Narrow the claim to call matching, public data, open 7B model, error flips	Kim et al. 2025 already tested roles and discussion in oncology	Kim et al. PDF text (VERIFIED)	Read their Appendix B myself
L-09	23 Oct 2026	Re-implement; do not copy MedAgents, MDAgents or Kim et al. code	Their repos have no licence	Repo file lists (VERIFIED)	Cite them as design references
L-10	23 Oct 2026	Conditional GO with the listed conditions and fallbacks	All required data and weights are openly licensed	Access matrix	Prompts 05–06

AI-use record

What the AI did	How it was checked	Outcome
Searched for neighbouring work (S1–S6)	Opened arXiv, ACL, NeurIPS and MLHC pages	Found Kim et al. 2025, which forced the claim to be narrowed (L-08)
Read Kim et al. 2025	Downloaded the PDF and searched its extracted text	Role-ablation and discussion numbers VERIFIED
Summarised MDAgents and MedAgentsBench	Summary of the PDF by a second model, not read in full	Labelled INFERRED; on my checklist
Tried to open the EvoMDT repository	git clone asked for login; page fetch failed	Marked UNKNOWN, not "unavailable"
Checked licences	Cloned MedAgents and MDAgents and listed files; fetched the Qwen model card	No licence on the reference repos; Qwen is Apache-2.0

What the AI did	How it was checked	Outcome
Estimated compute	Arithmetic from assumed tokens and throughput	INFERRED until the smoke test